

- Receives *MsgCounterSyncReq* message.
- Generates, encrypts and sends *MsgCounterSyncRsp* message.
- R1 and R2 each:
 - Receive, decrypt, process, and verify *MsgCounterSyncRsp* message from *Sender*.
 - On verification success: marks *Sender* 's group key message counter as synchronized.
 - Processes cached *Msg1* message.

4.19. Bluetooth Transport Protocol (BTP)

The Bluetooth Transport Protocol (BTP) provides a TCP-like layer of reliable, connection-oriented data transfer on top of GATT. BTP splits individual Service Data Unit (SDU) messages into multiple BTP segments, which are each transmitted via a single GATT write or indication (as shown in [Figure 22, “MATTERoBLE: Matter Message / BTP layering”](#)).

While BTP is a generic protocol, Matter specifically uses BTP to define a Matter-over-Bluetooth Low Energy (MATTERoBLE) Interface. A MATTERoBLE Interface SHALL implement BTP as a universally compatible transport mode. A MATTERoBLE Interface SHALL only be used to transport Matter messages as the BTP SDU.

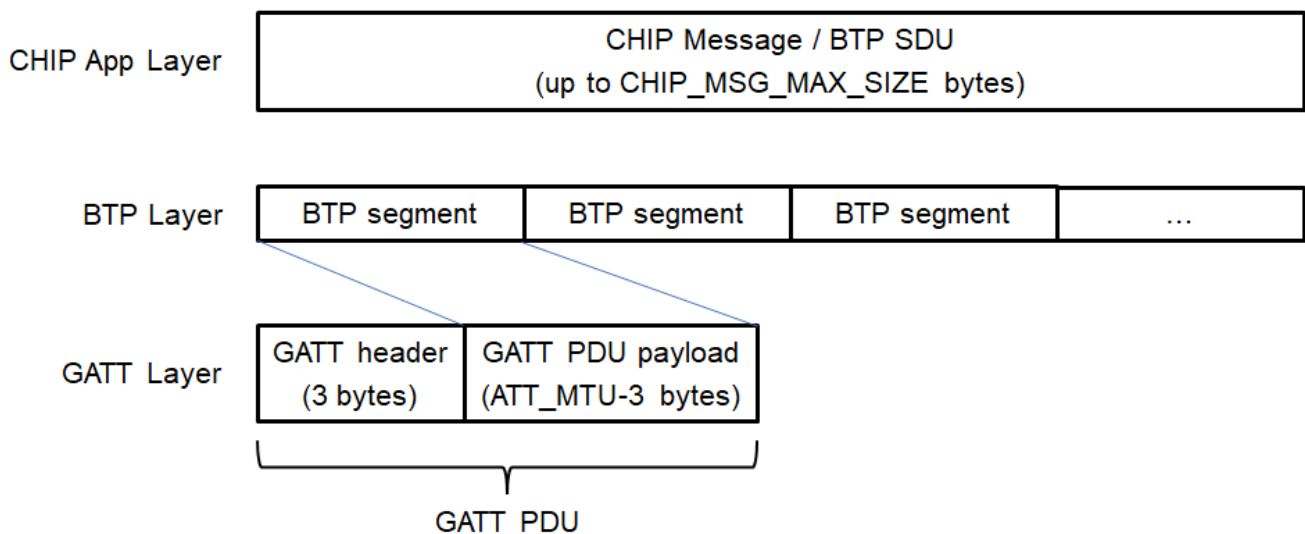


Figure 22. MATTERoBLE: Matter Message / BTP layering

The BTP session handshake allows devices to check BTP protocol version compatibility and exchange other data before a BTP session is established. Once established, this session is used to send and receive BTP SDUs (such as Matter messages) as BTP Message Segments. A BTP session MAY open and close with no effect on the state of the underlying Bluetooth LE connection, except in the case where a BTP session is closed by the Peripheral Device. Either the Peripheral or the Central MAY signal the end of a BTP session by closing the underlying BLE connection.

Due chiefly to constraints put on design by resource-limited BLE chipsets, BTP defines a receive window for each side of a session in units of GATT PDUs. Each GATT Write Characteristic Value (ATT_WRITE_REQ) PDU or Indication (ATT_HANDLE_VALUE_IND) PDU is sent with a sequence number which the receiver uses to acknowledge receipt of each packet at the BTP layer and open its receive window from the sender’s perspective.